

1 Specific Features of Machine Data Collection

The functions described in the paragraphs that follow are only active at workplaces meeting the following requirements:

- The machine/workplace has been configured as “individual workplace”
- The machine/workplace is assigned to a terminal, which has been configured in the “MDE” operation mode

These functions are not available at group workplaces.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

1.1 Shift automatic

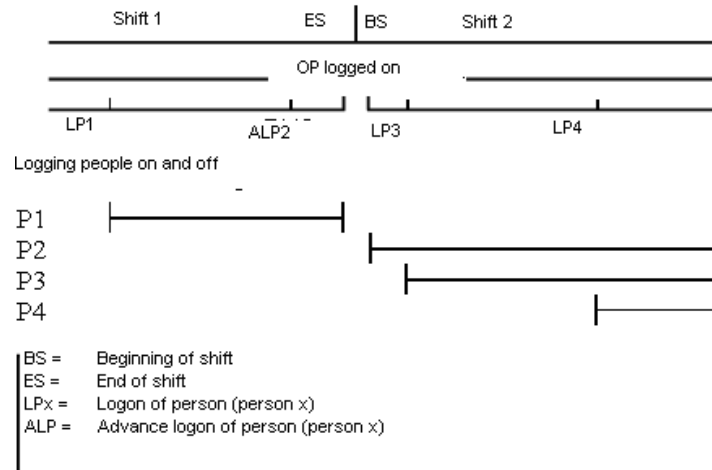
A shift model has to be assigned to each workplace/machine. Due to the information given by this shift calendar, the terminal is able to determine automatically the beginning and end of shifts for its assigned machines.

By the shift automatic option, functions are activated that ease data collection as well as operability:

- the OP that is logged on is automatically interrupted at the end of the shift.
- this OP is logged on at the beginning of the next shift
- Staff can log on in advance to a terminal within a certain period of time prior to the beginning of the shift. When the shift starts the next time, the terminal logs them on to the OP. This period of time is defined within the terminal configuration (option: waiting period for advance logon of staff).

Since data collection at terminals must not be interrupted and it is impossible for all terminals to send all postings simultaneously at the end of the shift, log records are buffered. The buffer is now transferred to the server in short intervals. This process depends on the number of machines defined for a terminal. Postings made during that time are buffered as well.

The following diagram shows a time flow of logons and logoffs during a change of shifts.



Description of the process

- 1) An OP was logged on during shift 1 and its production is also to be continued in the following shift 2.
- 2) Person1 logs on to the workplace.
- 3) Person 2 arrives shortly before the shift ends and logs on to the workplace. Since the logon takes place within the 30 minutes of advance logon time, the terminal recognizes an advance logon.
- 4) The registered OP is automatically interrupted when the shifts end and the staff logged on is logged off.
- 5) The OP, which was interrupted beforehand, is logged on again when the shift starts. Moreover, the persons who logged on in advance are logged on as well.

1.2 Downtime monitoring

The monitoring type is configured in the workplace/machine configuration (MDE configuration tab > Monitoring Type. The following values may be set:

Cyclic monitoring	Cycle time monitoring
Monitoring via operating signal	Operating signal monitoring
No monitoring	No automatic monitoring

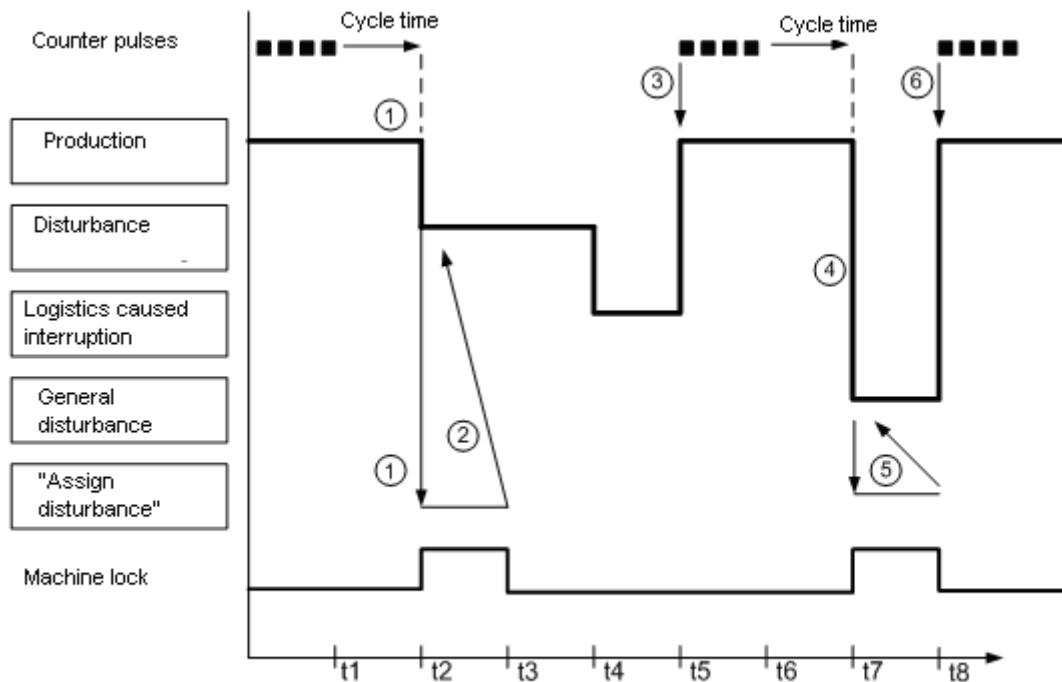
If, after a production phase (cycle time monitoring), the specified target cycle time is exceeded for the logged on order/OP, the terminal detects a downtime and expects a status to be entered. In this case the current terminal status is **"NOT ASSIGNED"**. The same principle applies to operation signal monitoring.

Monitoring of cycle time

When counter pulses arrive, the status is switched to "Production".

The cycle time is calculated from the target cycle of the OP multiplied by the value entered in the workplace/machine configuration (MDE Configuration tab > Monitoring type > Cycle extension").

A disturbance/malfunction can only be entered, if the AIP terminal requests it (no production).



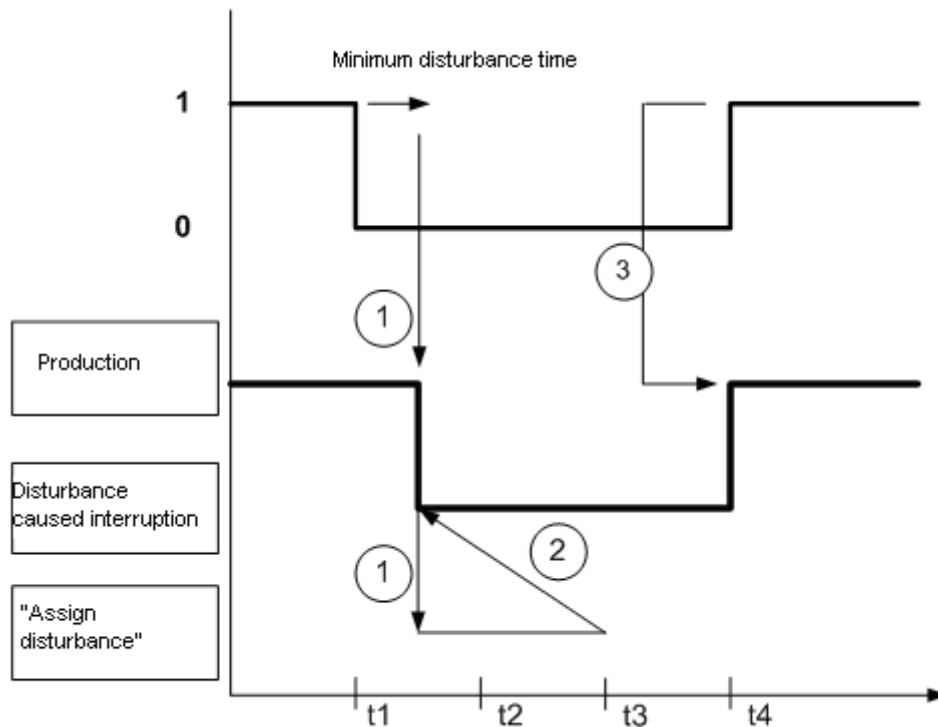
As a part of customizing, it is possible to activate a back posting for this terminal version, in order to record the point in time when the last pulse arrives for cycle time monitoring or the point in time when the operating signal stops for operating signal monitoring.

Operating signal monitoring

By setting the operating signal, status is changed to "Production".

A disturbance has to last for a certain time, before it will be identified as such and reported. This period is defined by "...MDE Configuration > Monitoring type > Minimum disturbance time".

A disturbance can only be entered, if the terminal requests it (no production).



As a part of customizing, it is possible to activate a back posting for this terminal version, in order to record the point in time when the last pulse arrives for cycle time monitoring or the point in time when the operating signal stops for operating signal monitoring.

No automatic monitoring

It is possible at all times to define a new machine status. The "Production" status must also be manually assigned.

1.3 Lock production status (production lock)

The "lock production status" button of the "workplaces/machines" section enables the user to prevent the terminal from switching automatically to the "production" status, when new clock pulses arrive from the machine. Thus, switching to "production" is disabled ("production lock").

In case a workplace/machine is unable to switch to the "production" status, the "note" column shows this by "production status locked".

In this case, the current status remains – despite the machine pulses received.

All items produced during the lock are either posted as yield, scrap or not at all, depending on the workplace configuration.

The production lock may be enabled or disabled explicitly by clicking the "lock production status" button.

If the workplace/machine status is configured accordingly (menu: master data > machines/workplaces > status assignment), the production lock may also be set automatically i.e. along with setting a status.

Authorization checking when setting the production lock manually

It is possible to allow the manual (explicit) setting or removing of the production lock only via a respective authorization.

For this purpose, the dynamic dialog M_PSPERRE is to be activated (customizing). The plant ID card (staff badge number) must be entered in this dialog.

If the dialog is active it will be opened, when the worker clicks the “lock production status” button. Once the badge number has been entered, it is checked whether this person is authorized to activate/remove the production lock. This is checked against the “change of production lock” authorization in the HR master (BDE tab).

The OFFLINE performance depends on the terminal configuration (“checking required” option).

Logging of the production lock

The manual activation/deactivation of the production lock is recorded as event at the server and may be evaluated within the machine history.

Please note in this context, that only the event of setting/removing the production lock manually will be recorded.

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But if the production lock is enabled or disabled implicitly by a status change, no explicit event will be recorded and, as a result it cannot be evaluated at the client.

In case a machine is locked for production and the AIP terminal is restarted, this production lock is automatically disabled after the restart. The process of changing the production lock is lock is neither recorded.

1.4 Machine lock

It can be defined for each status whether or not a “machine lock” is to be enabled as soon as this status is assigned. This can be configured in (status assignment > tab: control > other settings > set machine lock output).

Setting a machine lock leads to an output being set, which may trigger a relay, for example. In this case, the logical output is defined within the workplace/machine configuration (tab: MDE configuration > inputs/outputs > machine lock).

Notes on using the machine lock at DS 100 terminals

The value “1” has to be entered in the “machine lock” field of the configuration of machines/workplaces (tab: MDE configuration > inputs/outputs), provided that the relay is to be set at the DS100 terminal.

The relay is set for all statuses assigned to the “machine lock” as well as for the “not assigned” status.

1.5 Output “target quantity reached”

An output may be set to trigger a lamp, for example, via a relay (to be provided by the customer), when the target quantity of the currently registered operation has been reached.

The logical output is defined in the menu: master data > machines/workplaces > inputs/outputs > target quantity reached.

The target quantity of a machine is checked:

- once the terminal program *AIP* has been restarted
- after changing the target quantity using the corresponding posting dialog
- once an operation has been logged on, off or interrupted
- after posting manual quantities (partial uploads)
- after local quantity events of the MDE module (automatic quantities)
- once the order list has been reloaded

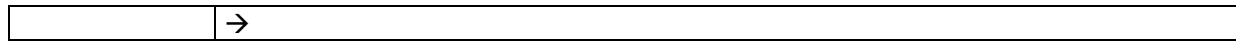
If several operations are logged on to a workplace/machine at the same time the logical output is set as soon as the target quantity has been reached for one of the orders. The signal will be reset if this OP is interrupted.

1.6 Scrap reasons depending on status & production lock

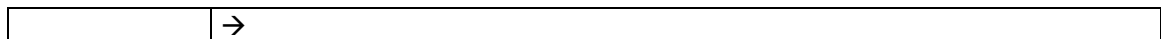
Two scrap reasons can be defined for a status. One scrap reason applying for the active production lock (see section **Error! Reference source not found.** Lock production status (production lock)) as well as one scrap reason applying for an inactive production lock. If a scrap reason is defined for the status that is currently available at the machine the counted scrap will be posted with this specified scrap reason.

Provided that a scrap reason is configured at the counting input, this one takes priority. Consequently, a counter that is assigned to a fixed scrap reason keeps this scrap reason even if another reason is configured for the currently available status.

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1.7 Setting of outputs subject to the status and posting scenarios

By way of an advanced configuration, digital outputs of the machine may be set if a specific machine status and a defined posting scenario coincide. Further details on this configuration can be found in the document entitled *MDE_digital_outputDepending_on_scenario.pdf*.

Please note the following for the local administration of the machine status at the terminal:

If status changes are posted using PDM or at another terminal the machine status is only transferred when reloading the machine list, provided that this has been configured explicitly by the "FollowExternStatus" option:

HYTNRCFG.INI
[Tnr Konfiguration 0] FollowExternStatus=on

When the status is changed by reloading the list, the terminal checks the configured logical outputs and sets or resets them, if required.